INFRASTRUCTURE INVESTMENT FEATURES

Learning Outcome

- explain features and characteristics of infrastructure

Infrastructure investments have a societal purpose; facilitate broad economic, technological, and social development purposes; and usually combine land, buildings, and other long-lived fixed assets. Infrastructure supports public transportation, airports, utilities (water, gas, and electricity), and more recently, information (telecommunication, cable, and wireless networks). Early infrastructure investments were typically financed by private corporations to realize profits. Subsequently, many governments took on a larger proportion of infrastructure investment and by the second half of the 20th century became the main source of investments. The late 20th century saw waves of privatizations of public transportation, and the scope of private infrastructure investment expanded to assets that governments had historically financed, owned, and operated themselves. The market for privately funded infrastructure is sizeable.

Infrastructure Investments

Infrastructure investments are real, capital-intensive, and long-lived assets intended for public use and provide essential services, such as airports, health care facilities, and sewage treatment plants. These investments have similarities to and differences from common equity and debt. Investments include equity, with its usual claims to residual cash flows, and debt, to finance and maintain such investments. Like real estate, investment in existing infrastructure involves acquiring unique, illiquid assets with distinct locations, features, and uses. Investments in construction and development of new infrastructure are made with expectations to generate cash either from income or from capital appreciation.

Infrastructure investments often involve a consortium that combines one or several strategic partners that have specialized operational or technical skills with the financial investors. Rather than leases or rentals from commercial or residential tenants, infrastructure cash flows in most cases arise from contractual payments, such as the following:

- *Availability payments*, which are payments that are received to make the facility available
- *Usage-based payments*, such as tolls and fees for using the facilities
- *"Take-or-pay" arrangements*, which obligate buyers to pay a minimum purchase price to sellers for a pre-agreed volume.

Allocations to infrastructure investments are driven both by the increased demand for infrastructure and by many governments' search for alternative funding sources for infrastructure investments, such as investors interested and experienced in building, managing, and running infrastructure. Investors have an interest in financing infrastructure investments, demand-side growth. As governments provide more opportunities by expanding infrastructure assets, they also continue to privatize government services.

Most infrastructure assets are financed, owned, and operated by governments, and a substantive proportion of these investments comes from public sources in the developing world. However, increasingly infrastructure is being financed privately through public–private partnerships (PPPs) by local, regional, and national governments. A **public–private partnership** is typically defined as a long-term contractual relationship between the public and private sectors for the purpose of having the private sector deliver a project or service traditionally provided by the public sector. Infrastructure investors may intend to

perhaps for even longer.

Infrastructure investments are also made in partnership with development finance institutions, which are specialized financial intermediaries that provide risk capital for economic development projects on a non-commercial basis. There are global, international, national, and local development finance institutions. For instance, the European Bank for Reconstruction and Development (EBRD) invests to improve municipal services, including infrastructure, and regularly taps global financial markets to finance infrastructure and other types of investments.

Categories of Infrastructure Investments

To categorize infrastructure investments, investors frequently rely on the underlying assets, with the broadest categorization distinguishing between economic and social infrastructure assets, as shown in Exhibit 6.

Exhibit 6:

Categorizing Infrastructure Investments

Economic Infrastructure Investments			
Transportation Assets	**Information and Communication Technology Assets**	**Utility and Energy Assets**	**Social Infrastructure Investments**
▫ roads ▫ bridges ▫ tunnels ▫ airports ▫ seaports ▫ railway systems	▫ telecommunication towers ▫ data centers	▫ electrical grid ▫ power generation, transmission, distribution ▫ potable water production ▫ gas storage and distribution ▫ liquefied natural gas terminals ▫ oil and gas infrastructure ▫ solid waste treatment	▫ educational assets ▫ health care assets ▫ social housing ▫ correctional facilities ▫ government/municipal buildings

Economic infrastructure investments support economic activity through transportation assets, information and communication technology (ICT) assets, and utility and energy assets:

- *Transportation assets* include roads, bridges, tunnels, airports, seaports, and heavy and light/urban railway systems. Income will usually be linked to demand based on traffic, airport and seaport charges, tolls, and rail fares and hence is deemed to carry market risk.
- *ICT assets* include infrastructure that stores, broadcasts, and transmits information or data, such as telecommunication towers and data centers.
- *Utility and energy assets* generate power and produce potable water; transmit, store, and distribute gas, water, and electricity; and treat solid waste.

Utility investments encompass environmentally sustainable development, with an increasing focus on renewable technologies, including solar, wind, and waste-to-energy power generation. Other energy assets may encompass downstream oil and gas infrastructure, the electrical grid, and liquefied natural gas terminals. The income earned from utility assets may also carry demand risk because buyers' energy and natural resources needs fluctuate. Alternatively, utilities can institute "take-or-pay" arrangements, locking buyers into minimum purchases whether supply is needed or not. Buyers usually have recourse if the utility falls short on performance or delivers supplies that are late or of inferior quality.

focus on providing, operating, and maintaining the asset infrastructure. The relevant services administered through those facilities are usually provided separately by the public authority or by a private service provider contracted by the public authority. In some countries, this model has been extended to other public infrastructure, including government and municipal buildings. Income from social infrastructure is typically derived from a type of lease payment that depends on availability payments and on managing and maintaining the asset according to predefined standards.

EXAMPLE 3

Clarkswood Infrastructure Fund LP

Clarkswood Infrastructure Fund LP invests its assets in equity and equity-like securities and debt issued by issuers that own or operate infrastructure assets in developed and developing countries. Infrastructure lays the foundation of basic services, facilities, and institutions on which the growth and development of a nation and country directly depend. More specifically, in several developed countries, such as the United States, the United Kingdom, and Australia, many infrastructure assets have been privatized as national, state, and local governments spend less on building, investing, maintaining, and operating infrastructure. The private sector has stepped in to provide equity financing to fund infrastructure.

For instance, the fund invests in an operator of Australian toll roads listed on the Australian Stock Exchange, as well as the operator of the Auckland Airport (New Zealand's largest airport), also listed on a stock exchange. It also invests in several corporations that provide infrastructure services in both the developing world and the developed world, including China and India, that see significant demand for infrastructure products and services and seek private sector investments. The fund invests up to 50% of its assets in US assets, up to 30% in infrastructure assets located in OECD (Organisation for Economic Co-operation and Development) countries, up to 30% in assets in non-OECD countries, and no more than 10% in any country with no more than 5% in any infrastructure provider.

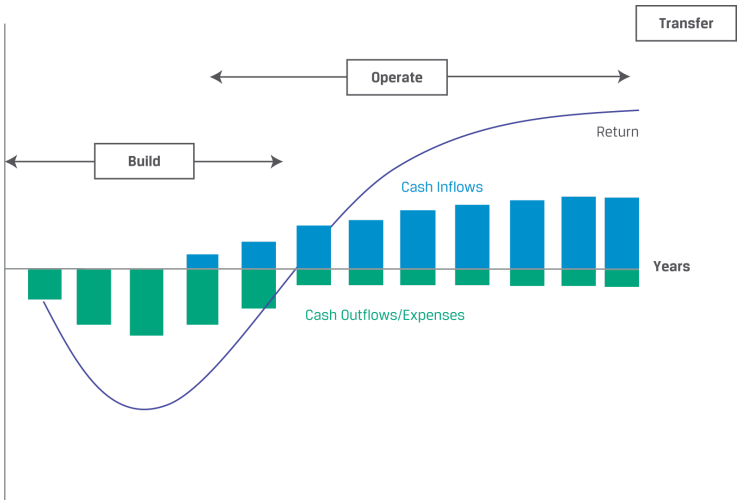
Stages of Infrastructure Development

Infrastructure investments can also be categorized by the underlying assets' stage of development. Typically, we distinguish among greenfield investments, secondary-stage investments, and brownfield investments.

Greenfield investments, developing new assets and new infrastructure, are opportunistic investments. The intent may be to lease or sell the assets to the government after construction or to hold and operate the assets. If they are held, it can be over the long term or for a shorter period until operational maturity, with subsequent sale to new investors, thus ensuring capital appreciation to reflect the construction and commissioning risk. Greenfield investors typically invest alongside strategic investors or developers that specialize in developing the underlying assets.

The greenfield investment life cycle common among public–private partnerships is called the build-operate-transfer (BOT) life cycle and is shown in Exhibit 7. The build phase is often characterized by an initial longer approval and construction phase with negative cash flows. The subsequent operate phase is governed by a concession agreement in which the private investor generates income based on pre-agreed parameters. In the final transfer phase, the investment is transferred to a government entity based on pre-determined parameters, sold to a third party, or decommissioned.

Exhibit 7:



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Brownfield investments expand existing facilities and may involve privatization of public assets or a sale leaseback of completed greenfield projects. They are characterized by a shorter investment period with immediate cash flows and an operating history. Typically, some of the assets' financial and operating history is available, and so brownfield investments may be sought after by both strategic investors specializing in operating the assets and—particularly with privatizations—financial investors focused on long-term, stable returns.

Secondary-stage investments invest in existing infrastructure facilities or fully operational assets that do not require further investment or development over the investment horizon. These assets generate immediate cash flow and returns expected over the investment period. Some assets never reach this stage because they continuously require additional capital and development.

Forms of Infrastructure Investment

As with real estate investments, infrastructure investments come in a variety of forms. The choice affects liquidity, cash flow, and income streams. Infrastructure investments can be direct or indirect. The vast majority of investors focus on equity investments, with some interest in pure debt (infrastructure bonds) and convertible debt.

- *Direct investment in the underlying infrastructure* provides control and the opportunity to capture full value. It requires a large investment and results in both concentration and liquidity risks while the assets are managed and operated. Because of this risk and the typical long-term horizon, direct infrastructure investment usually takes place with a group or consortium of strategic investors that share the financial risk and/or assume a specific role in building, operating, or managing the assets. Such strategic partners, such as large pension funds or sovereign wealth funds, are frequent direct investors because they are better placed to manage certain risks to limit individual concentration risk. Frequently these funds invest under specific mandates in infrastructure projects and specifically prioritize domestic infrastructure needs.

EXAMPLE 4

Public–Private Partnership, Iguá Saneamento S.A.

In March 2021, the Canada Pension Plan Investment Board paid CAD270 million (BRL1,178 million) for a 45% stake in Iguá Saneamento S.A. (Iguá), a water and sewage service company operating 18 concessions and contracts across five Brazilian states. The company serves more than 6 million people, provides sanitation

The objective of the transaction is to support Iguá's growth objectives and to maintain operating funds to support its operations. Iguá is considering acquiring additional water and sanitation concessions in Brazil, including the formation of public–private partnerships there. Other institutional investors include Alberta Investment Management Corporation (39%), BNDES Participações S.A. (11%), and IG4 Capital Group (11%).

• *Indirect investments* include infrastructure funds (similar in structure to private equity funds and either closed end or open end), infrastructure ETFs, and holding equity in publicly traded infrastructure providers, or master limited partnerships (MLPs). Investors concerned about liquidity and diversification may choose publicly traded infrastructure securities. Publicly traded infrastructure securities benefit from liquidity, reasonable fees, transparent governance, observable market prices, and transparent pricing, in addition to diversification among underlying assets. An investor should be aware, however, that publicly traded infrastructure securities represent a small segment of infrastructure investment: S&P Dow Jones Indices reported a total global market cap of USD2.23 trillion as of 31 December 2021.[2] Publicly traded infrastructure investments also tend to be clustered in certain asset categories.

Master limited partnerships (MLPs) trade on exchanges, are pass-through entities like REITs, and share the income pass-through structure taxation rules that minimize double taxation for investors. MLPs are most commonly used in energy transportation, processing, or storage; generate relatively stable cash flows from fee-based income; and distribute larger parts of their free cash flow to their investors.

Debt financing for infrastructure projects can be both private debt and publicly traded debt. Normally, the terms are flexible to accommodate periods of zero cash flow and long development or investment horizons. Publicly issued debt, such as the Airport Authority of Hong Kong perpetual bonds and Indonesian Infrastructure Fund US dollar bonds highlighted in earlier fixed-income modules are other approaches to financing infrastructure projects.

Question Set

1. Infrastructure cash flows primarily arise from:

A. dividends.
B. commercial tenants.
C. contractual payments.

Hide Solution

Solution:

The correct answer is C. Rather than leases or rentals from commercial or residential tenants, infrastructure cash flows arise from contractual payments, such as: availability payments (payments are received to make the facility available), usage-based payments (e.g., tolls and fees for using the facilities), and "take-or-pay" arrangements (which obligate buyers to pay a minimum purchase price to sellers for a pre-agreed volume).

2. Most infrastructure assets are financed, owned, and operated by:

A. governments.
B. public–private partnerships.
C. development finance institutions.

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The correct answer is A. Most infrastructure assets are financed, owned, and operated by governments, and a substantive proportion of these investments comes from public sources in the developing world. However, increasingly infrastructure is being financed privately through public–private partnerships by local, regional, and national governments. Infrastructure investments are also made in partnership with development finance institutions, which are specialized financial intermediaries that provide risk capital for economic development projects on a non-commercial basis.

3. Which of these statements about infrastructure investing is true? Infrastructure investments:

A. can generate cash income.
B. are intended to be non-profit.
C. do not have capital appreciation.

Hide Solution

Solution:

The correct answer is A. Investments in construction and development of new infrastructure are made with expectations to generate cash either from income or from capital appreciation.

4. Which of the following is a characteristic of direct investment in infrastructure?

A. High liquidity
B. Concentration risk
C. Short-term horizon

Hide Solution

Solution:

The correct answer is B. Direct investment in infrastructure requires a large investment and results in both concentration and liquidity risks while the assets are managed and operated. Because of these risks and the typical long-term horizon, direct infrastructure investment usually takes place with a group or consortium of strategic investors that share the financial risk and/or assume a specific role in building, operating, or managing the assets.

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